

Raspberry-Pi Based WAVEFORM GENERATOR

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A waveform generator is a commonly used test and measurement instrument in electronics laboratories. Waveforms of different functions, frequencies, and amplitude levels are often required to test the functionality of electronic circuits. Although the circuit presented is not a complete substitute for a laboratory-grade signal generator, it serves as a

waveform/signal generator with widely used standard waveforms, such as sinusoidal, square, triangular, and linear forward and reverse ramps under software control.

Moreover, since it is completely software-driven, it provides the added advantage of an 'arbitrary waveform generator' allowing users to design and generate custom signal waveforms with adjustable

rates (frequencies) or wave shapes.

The device uses the popular Raspberry Pi SBC as the main host controller, along with a few easily available, low-cost components that both hobbyists and professionals can assemble quickly.

The software is

written in Python for the Raspberry Pi. Fig. 1 shows the block diagram of the Raspberry Pi-based waveform generator. The components needed for this device are listed in Table 1.

Circuit and working

Fig. 2 illustrates the circuit diagram of this waveform generator. It is built around a Raspberry Pi, an 8-bit serial shift IC (74HC595), and a few other components.

The working of the circuit is straightforward. As shown in Fig. 2, the core of this waveform generator is the voltage-output R-2R DAC, comprising resistors R2 to R17. This 8-bit DAC accepts digital input and converts it directly into a voltage output. Unlike a current-output DAC, there is no need for an external voltage-to-current circuit using an op-amp. Thus, this is a very compact circuit that requires no additional power. Further details about the working and operation of this R-2R DAC can be found in the October 2000 issue of EFY, in the article 'R-2R DAC Based Waveform Generator Using PIC16C84 Micro-controller'. The output of this DAC is available at connector J1, which is labelled 'DAC Output'.

The digital data for this DAC input is provided by IC 74HC595, which is an 8-bit serial shift register. Data from the Raspberry Pi board is sent using the on-board SPI port, which uses the serial clock (SCLK) at GPIO pin 11. The serial data (SDATA) of the SPI port is available at GPIO pin 10 as master output slave input (MOSI).

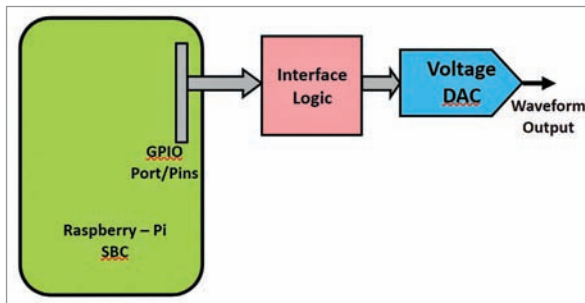


Fig. 1: Block diagram of the Raspberry Pi-based waveform generator

TABLE 1
BILL OF MATERIALS (COMPONENT LIST)

Name (Label)	Value	Comments
Raspberry-Pi	Board	For connecting GPIO pins
74HC595	IC	8-bit serial shift resistor
C1	100nF	Ceramic disc capacitor
D1	LED	5mm red/green
J1	2-pin header, male	Berg strip output connector
J2	3-pin header, male	Berg strip (optional); see text
J3	8-pin header, male	R-Pi interface connector
R1	1-kilo-ohm	0.25W, 5% CFR resistor
R3, R5, R7, R9, R11, R13, R15	10-kilo-ohm	0.25W, 1% MFR resistor
R2, R4, R6, R8, R10, R12, R14, R16, R17	20-kilo-ohm	0.25W, 1% MFR resistor
R18	4.7-kilo-ohm	Pull-up resistor (optional); see text
SW1	Tactile key 6mm	Key input (optional); see text
Jumper cables	10	Female-to-female jumper wires to connect the two boards and output